# JEE Mathematics: Quick Review with Grading and Practice Map - Circle (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (88 questions; a question that appears in two of these is counted once): High = 8 or more questions, Medium = 1 to 7 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

## Basic Geometry about Circles

### Question 01

What is the relation between the length of a chord and the angle subtended by the chord at the centre?

**Model Answer**

Chord subtending $2\theta$ at the centre: length $= 2r\sin\theta$; distance from centre $= r\cos\theta$. Angle at the circumference (major segment) $= \theta$.

Ex: $r = 3$, angle $\frac{2\pi}{3}$ at centre: chord $= 3\sqrt{3}$, mid-points of such chords lie on $x^2 + y^2 = \frac{9}{4}$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 20-21; J-II 22; J-M 20 | DPP: 2(4); 5(7, 10); 8(11, 15) | Question Bank: 53, 56

Also used in: Sheet: J-A 18 | DPP: 7(9)

### Question 02

Revise all properties mentioned in this heading.

**Model Answer**

Equal chords $\iff$ equal angles at centre $\iff$ equidistant from centre; nearer chord is longer. Angle at centre $= 2\times$ angle at circumference; angles in the same segment are equal; angle in a semicircle $= 90°$.

Cyclic quadrilateral: opposite angles sum to $180°$. Intersecting chords / secants: $AP{\cdot}PB = CP{\cdot}PD$. Alternate segment theorem. Tangents from a point are equal; the line to the centre bisects the angle between them and is the perpendicular bisector of the chord of contact.

**Grade** - Medium

Practice map, main concept: Sheet: J-II 4; J-M 23; C-I 4, 6, 12-13, 17-18, 22; C-II 18, 26 | DPP: 1(5, 12) | Question Bank: 49

Also used in: Sheet: J-I 20; J-M 20; C-I 15; C-II 15, 22 | DPP: 4(7)

## Circle: Definition and Equation

### Question 03

Define circle, its centre and radius.

**Model Answer**

Locus of a point in a plane whose distance from a fixed point (centre) is constant (radius).

**Grade** - Low

Practice map, main concept: none

Also used in: none

**Question 04**
How to find the equation of a circle if the centre and radius are known?

**Model Answer**

$$(x-a)^2+(y-b)^2=r^2$$

Ex: centre $(1,1)$ touching $3x+4y+18=0$: $r=\frac{25}{5}=5$, circle $x^2+y^2-2x-2y-23=0$.

**Grade** - **Medium**
Practice map, main concept: Sheet: J-I 2
Also used in: Sheet: J-II 15, 51; C-II 14 | DPP: 2(6) | Question Bank: 60, 64

**Question 05**
What is the general form of the equation of a circle?

**Model Answer**

$$x^2+y^2+2gx+2fy+c=0$$

Three independent constants $g, f, c$: three conditions fix a circle.

**Grade** - **Low**
Practice map, main concept: none
Also used in: none

**Question 06**
How to find the centre and radius from the general equation of a circle?

**Model Answer**
Make the coefficients of $x^2, y^2$ unity. Centre $(-g,-f)$ = $(-\frac{1}{2}$coeff of $x$, $-\frac{1}{2}$coeff of $y)$, $r=\sqrt{g^2+f^2-c}$.
Ex: $3x^2+3y^2-5x-6y+4=0 \Rightarrow$ centre $(\frac{5}{6},1)$, $r=\frac{\sqrt{13}}{6}$.

**Grade** - **High**
Practice map, main concept: Sheet: J-M 11-12, 31; C-I 20 | DPP: 1(1, 6-7, 13, 17-20) | Question Bank: 51
Also used in: Sheet: J-M 2, 16, 29; C-II 10 | Question Bank: 68

**Question 07**
What is the standard form of the equation of a circle?

**Model Answer**

$$x^2+y^2=r^2$$

Centre at the origin, radius $r$.

**Grade** - **Low**
Practice map, main concept: none
Also used in: none

**Question 08**

What is the necessary and sufficient condition for the general equation of 2nd degree to represent a circle?

**Model Answer**

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is a circle iff $a = b \neq 0$ and $h = 0$ (and, after dividing by $a$, $g^2 + f^2 - c \geq 0$ for a real circle).

**Grade** - Low

Practice map, main concept: DPP: 1(3)

Also used in: none

**Question 09**

What do you mean by point circle and imaginary circle?

**Model Answer**

$g^2 + f^2 - c > 0$: real circle; $= 0$: point circle (only the centre); $< 0$: imaginary circle (no real point).

**Grade** - Medium

Practice map, main concept: none

Also used in: Sheet: J-A 7 | DPP: 1(3)

**Question 10**

How to find the equation of a circle passing through 3 given points?

**Model Answer**

Put the points in $x^2+y^2+2gx+2fy+c = 0$ and solve for $g, f, c$; or take the centre $(a, b)$ equidistant from the points. A right angle at a point $\Rightarrow$ the opposite side is a diameter.

Ex: $(6, 0), (0, 8), (6, 8)$: centre $(3, 4)$, $r = 5$, circle $x^2 + y^2 - 6x - 8y = 0$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 1; J-II 14, 25, 57; J-M 16; J-A 14; C-II 2 | DPP: 2(13)

Also used in: Sheet: J-II 33 | DPP: 3(6)

**Question 11**

How to prove that 4 given points are concyclic?

**Model Answer**

Find the circle through three of them and check that the fourth satisfies it (or check opposite angles supplementary / $PA \cdot PB = PC \cdot PD$).

Ex: $(3, 4), (-3, -4), (0, 5), (-5, 0)$ all lie on $x^2 + y^2 = 25$.

**Grade** - Medium

Practice map, main concept: DPP: 1(10) | Question Bank: 38

Also used in: DPP: 1(12)

**Question 12**

How to find the image (reflection) of a circle in a given line?

**Model Answer**

Radius is unchanged; the centre goes to its image in the line: $\dfrac{x-h}{a} = \dfrac{y-k}{b} = \dfrac{-2(ah+bk+c)}{a^2+b^2}$.

Compare with the given form of the image (make coefficients of $x^2, y^2$ unity first).

Ex: image of $x^2 + y^2 + 16x - 24y + 183 = 0$ (centre $(-8, 12)$, $r = 5$) in $4x + 7y + 13 = 0$: centre $(-16, -2)$, circle $x^2 + y^2 + 32x + 4y + 235 = 0$.

**Grade** - Medium

Practice map, main concept: Sheet: J-M 35 | DPP: 1(2) | Question Bank: 40, 57, 60, 68

Also used in: none

## Diametrical Form of a Circle

**Question 13**

Write the expression for the diametrical form of a circle.

**Model Answer**

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

Ends of a diameter $A(x_1, y_1), B(x_2, y_2)$ ($\angle APB = 90°$). If $x_1, x_2$ are roots of $x^2 + px + q = 0$ and $y_1, y_2$ of $y^2 + ry + s = 0$, the circle is the sum of the two quadratics.

Ex: abscissae roots of $x^2+2x-a^2 = 0$, ordinates of $y^2+4y-b^2 = 0$: circle $x^2+y^2+2x+4y-a^2-b^2 = 0$.

**Grade** - Medium

Practice map, main concept: Sheet: J-M 22, 28, 30; C-II 22 | DPP: 1(4); 2(1-3, 5); 5(6, 13)

Also used in: Sheet: J-I 5; J-II 4, 47; J-M 23; J-A 8, 17; C-I 8, 22 | DPP: 3(12); 5(19); 8(13) | Question Bank: 57

**Question 14**

What is the smallest circle passing through 2 given points A and B?

**Model Answer**

The circle on $AB$ as diameter (centre = mid-point of $AB$, $r = \frac{AB}{2}$).

Ex: through $(1, 0), (0, 1)$: $x^2 + y^2 - x - y = 0$.

**Grade** - Low

Practice map, main concept: Sheet: J-I 3

Also used in: none

**Question 15**

How to find the equation of the circle drawn on the chord cut by a given line on a given 2nd degree curve as diameter?

**Model Answer**

Eliminate $y$: quadratic in $x$ (roots $x_1, x_2$); eliminate $x$: quadratic in $y$ (roots $y_1, y_2$). Make the leading coefficients equal and add. For a circle and a line use $S + \lambda L = 0$ with the centre on $L$.

Ex: $y = x$ on $x^2+y^2+10x-12y-40 = 0$: $x^2-x-20 = 0$, $y^2-y-20 = 0$; circle $x^2+y^2-x-y-40 = 0$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 26; J-II 3 | DPP: 8(13)

Also used in: DPP: 3(20) | Question Bank: 57

## Intercepts

**Question 16**

What is the formula for the length of a chord of a circle and how is it derived?

**Model Answer**

$$\text{chord} = 2\sqrt{r^2 - p^2}$$

$p$ = perpendicular distance of the chord from the centre; the perpendicular bisects the chord (Pythagoras).

Ex: $3x - 4y + 6 = 0$ on $x^2 + y^2 - 2x - 2y - 2 = 0$: $r = 2$, $p = 1$, chord $2\sqrt{3}$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 4; J-II 6-8, 69; J-M 6, 21; C-II 16 | DPP: 2(7); 8(10) | Question Bank: 42, 45, 69

Also used in: Sheet: J-I 20; J-II 22, 71; J-A 20

**Question 17**
What is the formula for the x-intercept and y-intercept of a circle?

**Model Answer**
$x$-intercept $= 2\sqrt{g^2 - c}$, $y$-intercept $= 2\sqrt{f^2 - c}$ (for $x^2 + y^2 + 2gx + 2fy + c = 0$).
Ex: touches $y$-axis at $(0, 4)$, $x$-intercept 6: centre $(\pm 5, 4)$, $r = 5$.

**Grade** - Medium
Practice map, main concept: Sheet: J-I 6; J-M 5, 29 | DPP: 2(6)
Also used in: Sheet: J-II 2, 63; J-A 4 | DPP: 2(2, 13, 16) | Question Bank: 41

**Question 18**
What is the general method to find the length of a chord for any 2nd degree curve?

**Model Answer**
Put $y = mx + c$ in the curve: quadratic in $x$ with roots $x_1, x_2$; chord $= |x_1 - x_2|\sqrt{1 + m^2} = \frac{\sqrt{D}}{|a|}\sqrt{1 + m^2}$.

**Grade** - Low
Practice map, main concept: Sheet: C-I 16
Also used in: none

**Question 19**
What is the general method to find the x-intercept and y-intercept for any 2nd degree curve?

**Model Answer**
Put $y = 0$ (or $x = 0$): intercept $= |x_1 - x_2| = \frac{\sqrt{D}}{|a|}$ of the resulting quadratic.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 20**
What is the condition for a circle to intersect the x-axis / y-axis at 2 distinct points?

**Model Answer**
$g^2 > c$ ($x$-axis); $f^2 > c$ ($y$-axis).

**Grade** - Medium
Practice map, main concept: Sheet: J-A 7
Also used in: none

**Question 21**
What is the condition for a circle to touch the x-axis / y-axis?

**Model Answer**
$g^2 = c$ ($x$-axis); $f^2 = c$ ($y$-axis).
Ex: through $(1, -2), (3, -4)$ touching the $x$-axis: $x^2+y^2-6x+4y+9 = 0$ or $x^2+y^2+10x+20y+25 = 0$.

**Grade** - Medium
Practice map, main concept: Sheet: J-II 67; J-A 4 | DPP: 2(15)
Also used in: Sheet: J-II 59; J-M 5; J-A 7 | Question Bank: 67

**Question 22**

What is the condition for a circle neither touching nor intersecting the x-axis / y-axis?

**Model Answer**

$g^2 < c$ (circle wholly above or below the $x$-axis); $f^2 < c$ (wholly left or right of the $y$-axis).

Ex: $x^2 + y^2 - 6x - 10y + c = 0$ meets neither axis and $(1, 4)$ inside: $25 < c < 29$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 9; J-M 1, 17 | DPP: 2(19)

Also used in: none

**Question 23**

What is the condition on the coordinates of the centre if the circle touches the x-axis / y-axis?

**Model Answer**

Touches $x$-axis: $|k| = r$; touches $y$-axis: $|h| = r$; touches both: $|h| = |k| = r$, i.e. $(x \mp r)^2 + (y \mp r)^2 = r^2$ (4 circles).

**Grade** - High

Practice map, main concept: Sheet: J-I 5; J-II 63 | DPP: 2(14, 16) | Question Bank: 41, 67

Also used in: Sheet: J-I 29; J-II 61; J-M 15, 21, 27; C-I 7; C-II 17, 27 | DPP: 3(16); 6(5); 7(6); 8(3) | Question Bank: 39, 46-47, 62, 69

**Question 24**

Draw the diagram of a chord drawn in a circle and find all lengths / area / angle.

**Model Answer**

Chord $AB$ at distance $p$, half-angle $\theta$ at centre: $AB = 2\sqrt{r^2 - p^2} = 2r\sin\theta$, $p = r\cos\theta$, area $\triangle OAB = \frac{1}{2}r^2 \sin 2\theta = p\sqrt{r^2 - p^2}$; angle at major arc $= \theta$, at minor arc $= \pi - \theta$.

**Grade** - Medium

Practice map, main concept: Sheet: J-II 5; J-M 32

Also used in: none

## Parametric Form of a Circle

**Question 25**

How to assume the coordinates of a point on a given circle?

**Model Answer**

$(h + r\cos\theta,\ k + r\sin\theta)$, $\theta \in [0, 2\pi)$, for centre $(h, k)$ and radius $r$; eliminate $\theta$ with $\cos^2\theta + \sin^2\theta = 1$ in locus questions.

Ex: $P$ on $x^2 + y^2 - 2x - 2y - 2 = 0$: mid-point of $OP$ lies on $x^2 + y^2 - x - y - \frac{1}{2} = 0$.

**Grade** - Medium

Practice map, main concept: Sheet: J-M 4; C-II 24 | DPP: 1(11); 2(9, 17) | Question Bank: 54

Also used in: Sheet: J-I 7; C-II 1 | Question Bank: 40

**Question 26**

What are the parametric coordinates of a point on the standard circle?

**Model Answer**

$(a\cos\theta,\ a\sin\theta)$ on $x^2+y^2=a^2$

Use for max / min of linear expressions: $\max(px+qy)=ph+qk+r\sqrt{p^2+q^2}$.

Ex: $(a,b)$ on $x^2+y^2-2x-2y-2=0$: $\max(5a-12b)=-7+26=19$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 8; J-II 9-10; C-I 11 | DPP: 2(10) | Question Bank: 44

Also used in: none

## Position of a Point w.r.t. a Circle

**Question 27**

How to define S, S1 and T for a point w.r.t. any 2nd degree curve?

**Model Answer**

$S$ = the curve's expression (RHS 0, coefficient of $x^2$ made 1 for a circle); $S_1=S(x_1,y_1)$; $T$: replace $x^2\to xx_1$, $y^2\to yy_1$, $2x\to x+x_1$, $2y\to y+y_1$, $2xy\to xy_1+yx_1$.

$$T\equiv xx_1+yy_1+g(x+x_1)+f(y+y_1)+c$$

**Grade** - Low

Practice map, main concept: none

Also used in: none

**Question 28**

How to find the relative position of a point w.r.t. a circle?

**Model Answer**

$S_1>0$: outside; $S_1=0$: on; $S_1<0$: inside (equivalently $CP>r,=r,<r$).

Ex: $(1,2)$: $S_1=10>0$ for $x^2+y^2-4x+6y-3$ (outside), $-6<0$ for $x^2+y^2+4x-6y-3$ (inside).

**Grade** - Medium

Practice map, main concept: Sheet: J-II 13, 54-55; J-M 8 | DPP: 2(12, 18) | Question Bank: 66

Also used in: Sheet: J-I 9; J-M 1, 17; C-II 21 | DPP: 1(19); 2(19); 5(17)

**Question 29**

How to find the greatest and least distance of a given point A from a circle?

**Model Answer**

$$\text{greatest}=AC+r,\quad \text{least}=|AC-r|$$

Ex: $M(-7,2)$, $x^2+y^2-10x-14y-151=0$ (centre $(5,7)$, $r=15$): $AC=13$, least distance 2.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 7; J-II 11; C-I 19 | Question Bank: 46

Also used in: none

**Question 30**

How to find the point on a circle lying at the greatest / least distance from a given point A?

**Model Answer**

Both lie on line $AC$: $C \pm r\,\hat{u}$ with $\hat{u}$ the unit vector along $CA$ (nearest on the side of $A$); use the parametric form of line $AC$.

Ex: farthest point of $x^2 + y^2 - 12x - 4y + 30 = 0$ from $O$: centre $(6, 2)$, $r = \sqrt{10}$, point $(9, 3)$.

**Grade** - Low

Practice map, main concept: Sheet: J-II 12

Also used in: none

**Question 31**

How to find the least distance between a line and a circle?

**Model Answer**

$p$ = distance of the line from the centre: least distance $= p - r$ if $p > r$, else 0.

Ex: $3x + 4y = 25$ and $x^2 + y^2 = 6x - 8y$ (centre $(3, -4)$, $r = 5$): $p = \frac{32}{5}$, least distance $\frac{7}{5}$.

**Grade** - Medium

Practice map, main concept: DPP: 3(2, 4)

Also used in: Sheet: J-II 18; J-M 9; C-I 9

## Line and Circle

**Question 32**

What is the condition of tangency for a circle and how is it used?

**Model Answer**

$$p = r$$

Distance of the line from the centre equals the radius. For $y = mx + c$ and $x^2 + y^2 = a^2$: $c^2 = a^2(1 + m^2)$. A circle touching two lines has its centre on an angle bisector.

Ex: $3x + 4y = c$ touches $x^2 + y^2 = 100$ iff $c = \pm 50$.

**Grade** - High

Practice map, main concept: Sheet: J-I 11; J-II 15, 19, 48-50, 58, 61, 68; J-M 14-15; J-A 1, 20; C-I 8, 10; C-II 15, 17 | DPP: 1(8, 14); 2(8); 3(1, 15-16, 20) | Question Bank: 47, 55, 59, 62, 64

Also used in: Sheet: J-I 10, 22; J-II 1, 8, 22, 53, 56; J-M 25; C-I 1-2; C-II 7, 12, 19 | DPP: 3(5, 10, 21); 6(16) | Question Bank: 58, 63

**Question 33**

How to determine the relative position of a line w.r.t. a circle?

**Model Answer**

$p > r$: no common point; $p = r$: tangent; $p < r$: secant; $p = 0$: diameter.

Ex: $3x + 4y = c$ cuts $x^2 + y^2 = 100$ at two points iff $-50 < c < 50$.

**Grade** - Medium

Practice map, main concept: Sheet: J-II 18, 56; J-M 9; C-II 21 | DPP: 1(9); 3(3, 5, 17); 7(12)

Also used in: Question Bank: 66

**Question 34**
What is the general method to find the relative position of a line w.r.t. a 2nd degree curve?

**Model Answer**
Eliminate one variable to get a quadratic: $D > 0$ secant, $D = 0$ tangent, $D < 0$ no meeting.

**Grade** - Low
Practice map, main concept: DPP: 2(11)
Also used in: Sheet: C-I 16

**Question 35**
What is the general method to find the condition of tangency of a line w.r.t. a 2nd degree curve?

**Model Answer**
Substitute the line in the curve and put the discriminant of the quadratic $= 0$.

**Grade** - Low
Practice map, main concept: none
Also used in: none

## Tangent and Normal

**Question 36**
Define tangent.

**Model Answer**
Limiting position of a secant when its two points of intersection coincide.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 37**
How to find the equation of the tangent to a 2nd degree curve if the point of tangency is given?

**Model Answer**
$T = 0$: $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$
Ex: tangent to $x^2 + y^2 = 5$ at $(1, -2)$: $x - 2y = 5$.

**Grade** - Medium
Practice map, main concept: Sheet: J-I 12; J-M 18; C-II 14
Also used in: Sheet: J-I 23; J-M 19, 30; J-A 1, 6

**Question 38**
How to find the equation of a tangent to a 2nd degree curve if any condition other than the point of tangency is given?

**Model Answer**
Assume the line ($y = mx + c$ with the given slope, or through the given point) and apply the condition of tangency $p = r$. For $x^2 + y^2 = a^2$: $y = mx \pm a\sqrt{1 + m^2}$.
Ex: tangents to $x^2 + y^2 - 6x + 4y = 12$ parallel to $4x + 3y + 5 = 0$: $4x + 3y - 31 = 0$, $4x + 3y + 19 = 0$.

**Grade** - Low
Practice map, main concept: Sheet: J-I 10; J-II 53 | DPP: 3(10, 18-19, 22)
Also used in: Sheet: J-I 4; J-II 14, 57 | DPP: 3(1)

**Question 39**
What is the equation of the tangent at a point given in parametric form on the standard circle?

**Model Answer**

$$x\cos\theta + y\sin\theta = a$$

At $(a\cos\theta, a\sin\theta)$ on $x^2+y^2=a^2$; its intercepts are $a\sec\theta$, $a\,\mathrm{cosec}\,\theta$.
Ex: mid-point of the intercepted part lies on $a^2(x^2+y^2) = 4x^2y^2$.

**Grade** - Medium
Practice map, main concept: Sheet: J-II 16-17; J-A 10; C-II 25 | DPP: 4(11)
Also used in: Sheet: J-A 6; C-I 13 | DPP: 7(13)

**Question 40**
How to find the equations of tangents to a circle passing through an external point?

**Model Answer**
Take $y - y_1 = m(x - x_1)$ and apply $p = r$: quadratic in $m$ gives two tangents (if one root is missing, the other tangent is $x = x_1$). Or use $SS_1 = T^2$ (67).
Ex: from $D(3,-3)$ to $(x+4)^2+(y+2)^2 = 25$: $3x - 4y = 21$ and $4x + 3y = 3$.

**Grade** - Low
Practice map, main concept: Sheet: J-II 20, 70 | DPP: 3(6, 13); 4(10)
Also used in: none

**Question 41**
Define normal.

**Model Answer**
The line through the point of contact perpendicular to the tangent.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 42**
How to find the equation of a normal to a circle?

**Model Answer**
Every normal passes through the centre: normal at $P$ = line $CP$; two given normals meet at the centre.
Ex: normal to $x^2+y^2 = 25$ through $(2,1)$: $x = 2y$.

**Grade** - Medium
Practice map, main concept: Sheet: J-II 51; J-M 25; J-A 6 | DPP: 3(8-9, 11-12, 14); 6(19)
Also used in: Sheet: J-M 24 | DPP: 6(3) | Question Bank: 55

**Question 43**
What is the equation of the chord joining two points $P(\alpha)$, $Q(\beta)$ on the standard circle?

**Model Answer**

$$x\cos\frac{\alpha+\beta}{2} + y\sin\frac{\alpha+\beta}{2} = a\cos\frac{\alpha-\beta}{2}$$

**Grade** - Low
Practice map, main concept: none
Also used in: none

## To Find the Point of Tangency

**Question 44**

How to find the point of tangency on a circle if the equation of the tangent is given?

**Model Answer**

Foot of the perpendicular from the centre to the tangent; or compare $T = 0$ (point $(x_1, y_1)$) with the given line.

Ex: $5x + 12y = 65$ on $x^2 + y^2 = 25$: $\frac{x_1}{5} = \frac{y_1}{12} = \frac{25}{65}$, point $(\frac{25}{13}, \frac{60}{13})$.

**Grade** - Medium

Practice map, main concept: Sheet: J-II 52; J-M 24 | DPP: 3(7, 21) | Question Bank: 63

Also used in: Sheet: J-I 12; J-A 20; C-II 24 | DPP: 3(15)

**Question 45**

How to find the point of tangency on a 2nd degree curve if the equation of the tangent is given?

**Model Answer**

Write the tangent at $(x_1, y_1)$ as $T = 0$ and compare coefficients with the given line (or solve line and curve: the double root).

**Grade** - Low

Practice map, main concept: none

Also used in: none

## Length of Tangent

**Question 46**

Write the expression for the length of the tangent from an external point.

**Model Answer**

$$L = \sqrt{S_1} = \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c}$$

Coefficients of $x^2, y^2$ must be 1 first.

Ex: from $(0, 5)$ to $x^2 + y^2 + 2x - 4 = 0$: $L = \sqrt{21}$; to $x^2 + y^2 - y + 1 = 0$: $L = \sqrt{21}$.

**Grade** - Low

Practice map, main concept: Sheet: J-I 13-15; J-II 62 | DPP: 4(2, 7, 14); 6(13)

Also used in: Sheet: J-II 20; C-II 13

**Question 47**

Define chord of contact for an external point.

**Model Answer**

The chord joining the points of contact of the two tangents from the point; it exists only if the point is not inside the circle.

**Grade** - Low

Practice map, main concept: none

Also used in: none

**Question 48**

How to find the angle between the tangents drawn from an external point to a circle?

**Model Answer**

If the angle between the tangents is $2\theta$: $\tan\theta = \dfrac{r}{L}$, $\sin\theta = \dfrac{r}{CP}$, so the angle $= 2\tan^{-1}\dfrac{r}{L}$.

Ex: from $O$ to $(x-7)^2+(y+1)^2=25$: $CP=\sqrt{50}$, $\sin\theta=\frac{1}{\sqrt{2}}$, angle $\frac{\pi}{2}$.

**Grade** - Low

Practice map, main concept: Sheet: J-I 17; C-II 13 | DPP: 4(9, 13); 5(14)

Also used in: none

**Question 49**

What is the angle subtended by a circle at an external point?

**Model Answer**

The angle between the two tangents from the point: $2\sin^{-1}\dfrac{r}{CP}$. Two circles subtend equal angles at $P$ iff $\frac{CP_1}{r_1}=\frac{CP_2}{r_2}$ (locus: a circle).

**Grade** - Low

Practice map, main concept: Sheet: C-I 14 | DPP: 6(12)

Also used in: none

**Question 50**

How to find the equation of the circumcircle of the triangle formed by the tangents from an external point and its chord of contact?

**Model Answer**

$PACB$ is cyclic with $CP$ as diameter: $(x-x_1)(x+g)+(y-y_1)(y+f)=0$.

Ex: $P(1,8)$, $x^2+y^2-6x-4y-11=0$: $(x-1)(x-3)+(y-8)(y-2)=0$, i.e. $x^2+y^2-4x-10y+19=0$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 16; J-II 26 | DPP: 4(4, 12)

Also used in: Sheet: J-A 13 | DPP: 8(20)

**Question 51**

Draw the diagram of tangents drawn from an external point to a circle and find all lengths / area / angle.

**Model Answer**

$L=\sqrt{S_1}$, $R=$ radius: area $PACB=RL$; chord of contact $AB=\dfrac{2RL}{\sqrt{R^2+L^2}}$; area $\triangle PAB=\dfrac{RL^3}{R^2+L^2}$; angle between tangents $\tan^{-1}\dfrac{2RL}{L^2-R^2}$.

Ex: $x^2+y^2=25$, $P(10,0)$: $L=5\sqrt{3}$, $PACB=25\sqrt{3}$, $AB=5\sqrt{3}$, $\triangle PAB=\frac{75\sqrt{3}}{4}$, angle $60°$.

**Grade** - Medium

Practice map, main concept: Sheet: J-M 3, 13, 33, 36; C-I 9, 24; C-II 1 | DPP: 4(5-6, 8)

Also used in: Sheet: J-II 20, 62; J-M 18-19

## Power of a Point

**Question 52**
Define the power of a point w.r.t. a circle and how to find it.

**Model Answer**

$$\text{Power} = S_1 = PT^2 = PA \cdot PB$$

Constant for every line through $P$ meeting the circle at $A, B$ (signed lengths along the line).
Ex: $P(5,2)$, $x^2+y^2=25$: $PA \cdot PB = 4$ for every secant.

**Grade** - Low
Practice map, main concept: Sheet: J-II 21; C-I 15; C-II 3 | DPP: 4(1, 3)
Also used in: Sheet: J-II 45, 62; C-I 24

**Question 53**
What is the power of a point according to its position relative to the circle?

**Model Answer**
Positive outside, zero on, negative inside the circle.

**Grade** - Low
Practice map, main concept: none
Also used in: none

## Director Circle

**Question 54**
Define the director circle of a curve.

**Model Answer**
Locus of the point of intersection of two perpendicular tangents to the curve.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 55**
What is the director circle of a circle?

**Model Answer**
$x^2+y^2=2a^2$ for $x^2+y^2=a^2$
Concentric, radius $\sqrt{2}\,r$. Ex: for $(x-2)^2+(y+1)^2=2$: $(x-2)^2+(y+1)^2=4$.

**Grade** - Low
Practice map, main concept: Sheet: C-II 4, 23
Also used in: Sheet: C-II 9

**Question 56**
What is the general method to find the director circle of a 2nd degree curve?

**Model Answer**
Write the tangent of slope $m$ (condition of tangency), pass it through $(h,k)$: quadratic in $m$; put $m_1 m_2 = -1$.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 57**
How to find the locus of an external point P for which the angle between the tangents from P to the circle is constant?

**Model Answer**
Angle $2\theta$: $CP = r\,\mathrm{cosec}\,\theta$, a concentric circle.

Ex: angle $60^\circ$ for $x^2 + y^2 = a^2$: $x^2 + y^2 = 4a^2$.

**Grade** - Low
Practice map, main concept: none
Also used in: none

## Chord whose Mid-point is Given

**Question 58**
Write the equation of a chord of a 2nd degree curve whose mid-point is given.

**Model Answer**

$$T = S_1$$

Locus of mid-points: write $T = S_1$ at $(h, k)$ and apply the given condition. Ex: chord of $x^2 + y^2 - 11x - 2y - 24 = 0$ bisected at $(1, 2)$: $9x - 2y - 5 = 0$.

**Grade** - Medium
Practice map, main concept: Sheet: J-II 23-24; J-A 3; C-I 5 | DPP: 5(3-5, 8-9, 11, 15, 17, 19) | Question Bank: 61
Also used in: DPP: 5(18)

**Question 59**
What is the formula for the length of a chord of a circle whose mid-point is given?

**Model Answer**

$$\text{length} = 2\sqrt{-S_1} = 2\sqrt{r^2 - CM^2}$$

Ex: circle $x^2 + y^2 - 4x - 6y + 8 = 0$ ($r^2 = 5$), mid-point $(1, 2)$: $CM^2 = 2$, chord $2\sqrt{3}$.

**Grade** - Medium
Practice map, main concept: Question Bank: 43
Also used in: none

**Question 60**
How to find the mid-point of a chord of a circle whose equation is given?

**Model Answer**
Foot of the perpendicular from the centre to the chord (the diameter perpendicular to the chord bisects it).

Ex: $2x - 5y + 18 = 0$ on $x^2 + y^2 - 6x + 2y - 54 = 0$: mid-point $(1, 4)$.

**Grade** - Medium
Practice map, main concept: Sheet: J-I 18-19; J-A 11 | DPP: 5(2, 16, 18)
Also used in: Question Bank: 42

**Question 61**
What is the general method to find the mid-point of a chord, whose equation is given, of a 2nd degree curve?

**Model Answer**
Take mid-point $(x_1, y_1)$, write $T = S_1$ and compare with the given chord; or solve line and curve and take $\frac{x_1+x_2}{2}$ from the sum of roots.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 62**
Which is the shortest chord of a circle passing through an internal point?

**Model Answer**
The chord bisected at that point (perpendicular to $CM$), length $2\sqrt{r^2 - CM^2}$; it is the chord through $M$ farthest from the centre. The longest is the diameter through $M$.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 63**
What is the locus of the mid-point of a variable chord of a circle passing through a fixed point P?

**Model Answer**
$\angle CMP = 90°$: the circle on $CP$ as diameter (the part inside the given circle).
Ex: chords of $x^2 + y^2 - 4x = 0$ through $(3, 4)$: $x^2 + y^2 - 5x - 4y + 6 = 0$.

**Grade** - Medium
Practice map, main concept: DPP: 5(1) | Question Bank: 52
Also used in: none

## Chord of Contact

**Question 64**
Write the equation of the chord of contact w.r.t. a 2nd degree curve for an external point.

**Model Answer**

$$T = 0$$

Ex: from $(8, 3)$ to $x^2 + y^2 = 49$: $8x + 3y = 49$.

**Grade** - Medium
Practice map, main concept: Sheet: J-I 22; C-II 19 | DPP: 5(12)
Also used in: Sheet: J-II 26; J-M 3; J-A 3 | DPP: 4(8); 5(15)

**Question 65**

How to find the external point if the equation of the chord of contact is given?

**Model Answer**

Write $T = 0$ for unknown $(x_1, y_1)$ and compare coefficients with the given line (the point is the pole of the line). The tangents at the ends of a chord meet at this point.

Ex: tangents at the ends of $x - 2y + 1 = 0$ on $x^2 + y^2 = 25$ meet at $(-25, 50)$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 23; J-II 27; J-M 19 | DPP: 5(20); 7(13-14)

Also used in: Sheet: J-I 25; J-II 25; J-A 9 | DPP: 8(15)

**Question 66**

What is the point of intersection of the tangents drawn at $P(\alpha)$, $Q(\beta)$ on the standard circle?

**Model Answer**

$$\left( \frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, \frac{a\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}} \right)$$

**Grade** - Medium

Practice map, main concept: Sheet: J-A 9 | Question Bank: 58

Also used in: none

## Pair of Tangents

**Question 67**

Write the combined equation of the pair of tangents from an external point to a 2nd degree curve.

**Model Answer**

$$SS_1 = T^2$$

Ex: from the origin to $x^2 + y^2 + 2gx + 2fy + c = 0$: $(gx + fy)^2 = c(x^2 + y^2)$.

**Grade** - Low

Practice map, main concept: Sheet: J-I 24-25

Also used in: Sheet: J-II 70

## Family of Circles: Type 1

**Question 68**

How to find the equation of a circle passing through the points of intersection of two circles S1 and S2?

**Model Answer**

$S_1 + \lambda S_2 = 0$ $(\lambda \neq -1)$, or $S_1 + \lambda(S_1 - S_2) = 0$

Find $\lambda$ from the extra condition. Ex: through the meet of $x^2 + y^2 - 6x + 2y + 4 = 0$, $x^2 + y^2 + 2x - 4y - 6 = 0$ with centre on $y = x$: $7x^2 + 7y^2 - 10x - 10y - 12 = 0$.

**Grade** - Low

Practice map, main concept: Sheet: J-I 35; J-II 32, 40, 46; C-I 3 | DPP: 6(15, 17-18); 8(20)

Also used in: none

**Question 69**
How to check whether the value of $\lambda$ is infinite by observing the calculations, and how to find the required circle in that case?

**Model Answer**
If the condition gives an impossible equation in $\lambda$ (e.g. $0 = k$) or is already satisfied by $S_2$, then $\lambda \to \infty$: the answer is $S_2 = 0$ itself. Writing $\mu S_1 + S_2 = 0$ avoids the loss.

**Grade** - Low
Practice map, main concept: none
Also used in: Sheet: J-I 35

## Family of Circles: Type 2

**Question 70**
How to find the equation of a circle passing through the points of intersection of a circle S and a line L?

**Model Answer**

$$S + \lambda L = 0$$

Ex: through $(3, 4)$ and the meet of $x^2 + y^2 = 9$, $x + y = 3$: $\lambda = -4$, $x^2 + y^2 - 4x - 4y + 3 = 0$.

**Grade** - Low
Practice map, main concept: DPP: 6(16)
Also used in: Sheet: J-I 26

**Question 71**
What does the equation of a variable circle $S + \lambda L = 0$ represent?

**Model Answer**
Circles through the two fixed points where $S = 0$ meets $L = 0$. To find them, group the $\lambda$ terms: solve $S = 0$, $L = 0$.
Ex: $x^2 + y^2 + kx + (1 + k)y - (k + 1) = 0$ passes through $(1, 0)$ and $(\frac{1}{2}, \frac{1}{2})$.

**Grade** - Low
Practice map, main concept: Sheet: J-II 60; C-II 10 | DPP: 8(14)
Also used in: Sheet: J-II 41 | DPP: 4(3)

## Family of Circles: Type 3

**Question 72**
How to find the equation of the family of circles passing through two given points A and B?

**Model Answer**

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + \lambda \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Circle on $AB$ as diameter $+\lambda$ (line $AB$). Centres lie on the perpendicular bisector of $AB$.

**Grade** - Low
Practice map, main concept: Sheet: J-II 33, 59
Also used in: Sheet: J-II 67; C-II 8 | DPP: 2(15)

## Family of Circles: Type 4

**Question 73**

How to find the equation of a circle touching a given line L at its given point P?

**Model Answer**

$$(x - x_1)^2 + (y - y_1)^2 + \lambda L = 0$$

Ex: through $O$, touching $x + y = 2$ at $(1, 1)$: $\lambda = 1$, $x^2 + y^2 - x - y = 0$.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 27-28; J-II 29, 44, 47; J-M 7 | DPP: 6(1-3); 8(4)

Also used in: Sheet: C-II 9

**Question 74**

How to find the equation of a circle touching a given 2nd degree curve at its given point P?

**Model Answer**

Use the tangent at $P$ $(T = 0)$: $(x - x_1)^2 + (y - y_1)^2 + \lambda T = 0$; for a given circle $S$ also $S + \lambda T = 0$.

Ex: through $(1, 1)$, touching $x^2 + y^2 + 4x - 6y - 3 = 0$ at $(2, 3)$: $x^2 + y^2 + x - 6y + 3 = 0$.

**Grade** - Low

Practice map, main concept: Sheet: J-II 30

Also used in: Sheet: J-II 31

**Question 75**

How to find the equation of the circumcircle of the triangle formed by three lines whose equations are given?

**Model Answer**

$$L_1L_2 + \lambda L_2L_3 + \mu L_3L_1 = 0$$

Choose $\lambda, \mu$ so that coeff of $x^2$ = coeff of $y^2$ and coeff of $xy = 0$. Ex: $x = 0$, $y = 0$, $x + y = 4$: $x^2 + y^2 - 4x - 4y = 0$.

**Grade** - Low

Practice map, main concept: DPP: 1(21)

Also used in: none

**Question 76**

How to find the equation of the circumcircle of a cyclic quadrilateral formed by 4 lines whose equations are given?

**Model Answer**

$$L_1L_3 + \lambda L_2L_4 = 0$$

Sides in order $L_1, L_2, L_3, L_4$; fix $\lambda$ by coeff of $x^2$ = coeff of $y^2$, coeff of $xy = 0$. Ex: $2x + 3y = 2$, $3x - 2y = 4$, $x + 2y = 3$, $2x - y = 3$ in this order: $8x^2 + 8y^2 - 25x - 3y + 18 = 0$.

**Grade** - Low

Practice map, main concept: Sheet: J-II 1

Also used in: none

## Common Tangents

**Question 77**
Define common tangents and their types DCT and TCT.

**Model Answer**
A line touching both circles. Direct common tangent (DCT): both circles on the same side; transverse common tangent (TCT): circles on opposite sides.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 78**
Which common tangent is called external and which one internal?

**Model Answer**
DCT = external common tangent; TCT = internal (transverse) common tangent (it passes between the circles).

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 79**
How to check whether a common tangent is a DCT or a TCT?

**Model Answer**
Put both centres in the line $L$: same sign $\Rightarrow$ DCT, opposite signs $\Rightarrow$ TCT.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 80**
What is the relative position of the centres of two circles w.r.t. a DCT / TCT?

**Model Answer**
Both centres on the same side of a DCT; on opposite sides of a TCT.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 81**
In what ratio does the point of intersection of the DCTs / TCTs divide the line joining the centres?

**Model Answer**
DCTs meet at $P$ dividing $C_1C_2$ externally in $r_1 : r_2$; TCTs meet at $R$ dividing it internally in $r_1 : r_2$. Ex: $C_1C_2 = 35$, $r_1 = 12$, $r_2 = 9$: $C_1R = 35 \cdot \frac{12}{21} = 20$.

**Grade** - Medium
Practice map, main concept: Sheet: J-A 12 | DPP: 6(6); 7(1, 8)
Also used in: DPP: 7(15) | Question Bank: 50

**Question 82**
What is the condition on the DCTs if the radii of the two circles are equal?

**Model Answer**
They are parallel to the line joining the centres (no finite meeting point).

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 83**
What is the formula for the length of a DCT / TCT?

**Model Answer**

$$L_{DCT} = \sqrt{d^2 - (r_1 - r_2)^2}, \qquad L_{TCT} = \sqrt{d^2 - (r_1 + r_2)^2}$$

Ex: $x^2 + y^2 - 20x + 64 = 0$, $x^2 + y^2 + 30x + 144 = 0$: $d = 25$, $r_1 = 6$, $r_2 = 9$: $L_{TCT} = 20$.

**Grade** - Medium
Practice map, main concept: Sheet: J-II 34 | DPP: 6(9, 11)
Also used in: Sheet: J-II 65, 71; J-A 12, 19; C-I 23

**Question 84**
How to find the equations of the DCTs / TCTs of two circles?

**Model Answer**
Find $P$ (external) / $R$ (internal division in $r_1 : r_2$), then tangents from that point to either circle ($y - k = m(x - h)$, $p = r$). If $r_1 = r_2$, the DCTs are $y = mx + c$ with $m =$ slope of $C_1C_2$.
Ex: $x^2+y^2 = 1$, $(x-1)^2+(y-3)^2 = 4$: common tangents $y = 1$, $x+1 = 0$, $3x+4y = 5$, $4x-3y = 5$.

**Grade** - Medium
Practice map, main concept: Sheet: J-A 2; C-I 1 | DPP: 6(10); 7(18)
Also used in: Sheet: C-II 16

**Question 85**
How to find the equation of the DCT / TCT if two circles touch each other?

**Model Answer**
The common tangent at the point of contact is $S_1 - S_2 = 0$; the point of contact divides $C_1C_2$ in $r_1 : r_2$ (internally if they touch externally, externally if internally).
Ex: $x^2 + y^2 - 4x - 6y - 12 = 0$, $x^2 + y^2 + 6x + 18y + 26 = 0$: $5x + 12y + 19 = 0$.

**Grade** - Medium
Practice map, main concept: Sheet: J-M 10 | DPP: 7(15)
Also used in: Sheet: J-M 34

**Question 86**
How to find the angle between the DCTs / TCTs?

**Model Answer**
Half-angle $\theta$: $\sin\theta = \dfrac{r_1 - r_2}{d}$ for DCTs, $\sin\theta = \dfrac{r_1 + r_2}{d}$ for TCTs; the angle is $2\theta$.
Ex: radii 4 and 1 touching externally ($d = 5$): $\sin\theta = \frac{3}{5}$, $\sin 2\theta = \frac{24}{25}$.

**Grade** - Low
Practice map, main concept: DPP: 6(4)
Also used in: none

## Position of Two Circles relative to Each Other

**Question 87**

When are two circles separated from each other, and how many common tangents are there?

**Model Answer**

$d > r_1 + r_2$: 4 common tangents (2 DCT, 2 TCT).

**Grade** - Medium

Practice map, main concept: Sheet: J-II 65, 72 | DPP: 7(5)

Also used in: Sheet: J-A 15-16

**Question 88**

When do two circles touch each other externally, and how many common tangents are there?

**Model Answer**

$d = r_1 + r_2$: 3 common tangents. A circle touching a fixed circle externally and a fixed line: distance to centre $= r + R$, distance to line $= r$ (locus: parabola).

Ex: $x^2 + y^2 - 2x - 4y + 1 = 0$, $x^2 + y^2 + 4x + 4y - 1 = 0$: $d = 5 = 2 + 3$, touch externally.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 29, 31, 37; J-II 2, 36, 64, 66; J-M 27; J-A 18; C-I 7; C-II 5, 27 | DPP: 6(5, 7); 7(7, 10-11, 16-17) | Question Bank: 39, 50

Also used in: Sheet: J-II 72; C-I 10, 21 | DPP: 3(11); 6(19)

**Question 89**

When do two circles intersect at 2 distinct points, and how many common tangents are there?

**Model Answer**

$|r_1 - r_2| < d < r_1 + r_2$: 2 common tangents (DCTs only).

Ex: $(x + 1)^2 + (y + 2)^2 = r^2$ and $x^2 + y^2 - 4x - 4y + 4 = 0$ ($d = 5$, $r_2 = 2$): $3 < r < 7$.

**Grade** - Low

Practice map, main concept: Sheet: J-II 35; C-I 23 | DPP: 7(2)

Also used in: Sheet: J-II 72

**Question 90**

When does one circle touch another internally, and how many common tangents are there?

**Model Answer**

$d = |r_1 - r_2|$: 1 common tangent.

**Grade** - Medium

Practice map, main concept: Sheet: J-I 30; J-II 31; J-M 34; J-A 17; C-I 21; C-II 12 | DPP: 6(8); 7(6)

Also used in: Sheet: J-II 72; C-II 27

**Question 91**

When is one circle contained in another, and how many common tangents are there?

**Model Answer**

$d < |r_1 - r_2|$: no common tangent.

**Grade** - Medium

Practice map, main concept: Sheet: J-M 2; J-A 15-16

Also used in: DPP: 7(5)

**Question 92**
What is the minimum distance between two circles in every relative position?

**Model Answer**
Separate: $d-(r_1+r_2)$; touching or intersecting: 0; one inside the other: $|r_1-r_2|-d$. Maximum distance: $d+r_1+r_2$.

**Grade** - Low
Practice map, main concept: none
Also used in: DPP: 6(11)

## Radical Axis

**Question 93**
Define the radical axis of two circles.

**Model Answer**
Locus of a point whose powers (squares of tangent lengths) w.r.t. the two circles are equal.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 94**
Write the expression for the radical axis of two circles S1 = 0 and S2 = 0.

**Model Answer**

$$S_1 - S_2 = 0: \ 2(g_1-g_2)x + 2(f_1-f_2)y + c_1 - c_2 = 0$$

Coefficients of $x^2, y^2$ must be 1 in both.

**Grade** - Medium
Practice map, main concept: Sheet: J-II 37; C-II 7 | DPP: 7(3)
Also used in: Sheet: J-I 32-33; J-II 39; J-A 19 | DPP: 7(14)

**Question 95**
Which line is the radical axis of two circles if they intersect at 2 distinct points?

**Model Answer**
Their common chord.

**Grade** - Low
Practice map, main concept: none
Also used in: Sheet: J-II 37 | DPP: 7(3)

**Question 96**
Which line is the radical axis of two circles if they touch each other?

**Model Answer**
The common tangent at the point of contact.

**Grade** - Low
Practice map, main concept: Sheet: J-I 32
Also used in: Sheet: J-II 64, 66

**Question 97**
What is the condition on the radical axis w.r.t. the line joining the centres?

**Model Answer**
It is perpendicular to $C_1C_2$ (but need not pass through its mid-point).

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 98**
What is the condition on the radical axis w.r.t. a common tangent of two circles?

**Model Answer**
It bisects every common tangent (equal tangent lengths from the meeting point).

**Grade** - Medium
Practice map, main concept: Sheet: J-A 8, 19
Also used in: Sheet: J-II 34

**Question 99**
Which type of two circles will not have a radical axis?

**Model Answer**
Concentric circles ($S_1 - S_2 = c_1 - c_2 \neq 0$ is not a line).

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 100**
How to find the equation of the common chord of two circles?

**Model Answer**

$$S_1 - S_2 = 0$$

Length $= 2\sqrt{r_1^2 - p_1^2}$, $p_1 =$ distance of $S_1 - S_2 = 0$ from $C_1$; circle on it as diameter: $S_1 + \lambda(S_1 - S_2) = 0$ with centre on the chord.
Ex: radii 4 and 8 cutting orthogonally: $d = 4\sqrt{5}$, common chord $= \frac{2r_1r_2}{d} = \frac{16}{\sqrt{5}}$.

**Grade** - Medium
Practice map, main concept: Sheet: J-II 71; C-II 20 | DPP: 7(9) | Question Bank: 48
Also used in: Sheet: C-I 3 | DPP: 6(17); 7(4); 8(6) | Question Bank: 65

**Question 101**
What is the condition if one circle bisects the circumference of the other circle?

**Model Answer**
If $S_2$ bisects the circumference of $S_1$, the common chord $S_1 - S_2 = 0$ passes through the centre of $S_1$ (it is a diameter of $S_1$): $r_2^2 = r_1^2 + C_1C_2^2$.
Ex: $x^2 + y^2 - 2x + 3y + c = 0$ bisects $x^2 + y^2 = 9$: $c = -9$.

**Grade** - Medium
Practice map, main concept: Sheet: J-I 33-34, 36; J-II 38; J-M 26 | DPP: 6(14); 7(4)
Also used in: Sheet: J-M 7 | DPP: 1(15)

**Question 102**
How to check the relative position of two circles with the help of the radical axis?

**Model Answer**
Compare the distance $p$ of $S_1 - S_2 = 0$ from $C_1$ with $r_1$: $p < r_1$ intersect, $p = r_1$ touch, $p > r_1$ no common point (separate or one inside the other; the axis then lies outside both).

**Grade** - Low
Practice map, main concept: none
Also used in: none

## Radical Centre

**Question 103**
Define the radical centre of three circles.

**Model Answer**
The common point of the three radical axes taken in pairs (exists if the centres are not collinear).

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 104**
How to find the radical centre of three circles if their equations are given?

**Model Answer**
Solve any two of $S_1 - S_2 = 0$, $S_2 - S_3 = 0$ (coefficients of $x^2, y^2$ unity).
Ex: $x^2 + y^2 - 3x - 6y + 14 = 0$, $x^2 + y^2 - x - 4y + 8 = 0$, $x^2 + y^2 + 2x - 6y + 9 = 0$: radical centre $(1, 2)$.

**Grade** - Low
Practice map, main concept: Sheet: C-II 8 | DPP: 1(15); 8(17)
Also used in: none

**Question 105**
Which is the point from which the tangents drawn to the three circles are equal?

**Model Answer**
The radical centre.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 106**
Which point is the radical centre of three circles drawn on the three sides of a triangle as diameters?

**Model Answer**
The orthocentre of the triangle (each radical axis is an altitude).

**Grade** - Low
Practice map, main concept: none
Also used in: none

## Coaxial System of Circles

**Question 107**
Define a coaxial system of circles.

**Model Answer**
A family of circles every two of which have the same radical axis.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 108**
How to find the equation of a circle belonging to a given coaxial system?

**Model Answer**
$S_1 + \lambda(S_1 - S_2) = 0$, i.e. $S + \lambda L = 0$ ($L$ = radical axis)
Ex: through $O$, coaxial with $x^2+y^2 = 9$ and $x^2+y^2-6x+3 = 0$: $x^2+y^2-9+3(x^2+y^2-6x+3) = 0$, i.e. $2x^2 + 2y^2 - 9x = 0$.

**Grade** - Low
Practice map, main concept: Sheet: J-II 39; C-II 9
Also used in: none

## Angle between a Line and a Circle

**Question 109**
Define the angle of intersection of two curves.

**Model Answer**
The angle between their tangents at a common point.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 110**
What is the angle of intersection of two curves if they have no common point?

**Model Answer**
Not defined.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 111**
How to find the angle between a line and a circle?

**Model Answer**

$$\cos\theta = \frac{p}{r}$$

$p$ = distance of the line from the centre ($\theta = 0$ for a tangent). Ex: $x^2 + y^2 = 2$ and $3x + 4y = 5$: $p = 1$, $\cos\theta = \frac{1}{\sqrt{2}}$, $\theta = 45°$.

**Grade** - Low
Practice map, main concept: DPP: 8(1, 12)
Also used in: none

**Question 112**
What do you mean when a line cuts a circle orthogonally?

**Model Answer**
$\theta = 90°$: $p = 0$, the line passes through the centre (it is a normal). A line pair cuts a circle orthogonally iff both lines pass through the centre.

**Grade** - Low
Practice map, main concept: Sheet: J-II 42
Also used in: Sheet: J-II 43 | DPP: 8(5)

## Angle between Two Circles and Orthogonality

**Question 113**
How to find the angle between two circles?

**Model Answer**

$$\cos\theta = \frac{r_1^2 + r_2^2 - d^2}{2r_1 r_2}$$

Ex: $(x-1)^2 + y^2 = 10$, $x^2 + (y-2)^2 = 5$: $d^2 = 5$, $\cos\theta = \frac{10}{2\sqrt{50}} = \frac{1}{\sqrt{2}}$, $\theta = \frac{\pi}{4}$.

**Grade** - Medium
Practice map, main concept: DPP: 8(9) | Question Bank: 65
Also used in: none

**Question 114**
What is the condition of orthogonality of two circles in terms of their radii and the distance between centres?

**Model Answer**

$$r_1^2 + r_2^2 = d^2$$

Ex: equal circles centred $(2,3)$, $(5,6)$ cutting orthogonally: $2r^2 = 18$, $r = 3$.

**Grade** - Low
Practice map, main concept: DPP: 8(2-3, 6)
Also used in: none

**Question 115**
What is the condition of orthogonality of two circles in terms of their equations?

**Model Answer**

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

Equivalent: tangent to one at a common point passes through the other's centre; the power of $C_1$ w.r.t. $S_2$ is $r_1^2$.
Ex: $x^2 + y^2 - 2x - 4 = 0$ and $x^2 + y^2 + 16x + 14y - 12 = 0$: $2(-1)(8) + 0 = -16 = c_1 + c_2$.

**Grade** - Medium
Practice map, main concept: Sheet: J-I 39; J-II 41, 43, 45; J-A 5; C-I 2 | DPP: 8(5, 7, 16, 18)
Also used in: Sheet: J-II 40, 42, 44, 46-47 | DPP: 8(4, 8)

**Question 116**
How to find the equation of a circle cutting three given circles orthogonally?

**Model Answer**
Centre = radical centre of the three circles; radius = tangent length from it to any of them (or solve three orthogonality equations for $g, f, c$).
Ex: $x^2+y^2-2x+3y-7=0$, $x^2+y^2+5x-5y+9=0$, $x^2+y^2+7x-9y+29=0$: $x^2+y^2-16x-18y-4=0$.

**Grade** - Low
Practice map, main concept: Sheet: J-I 38 | DPP: 8(19)
Also used in: none

**Question 117**
What is the locus of the centre of a variable circle cutting two given circles orthogonally?

**Model Answer**
The radical axis of the two circles (part outside them).
Ex: $x^2+y^2+4x-6y+9=0$, $x^2+y^2-5x+4y+2=0$: $9x-10y+7=0$.

**Grade** - Medium
Practice map, main concept: Sheet: C-II 11 | DPP: 8(8)
Also used in: Sheet: J-A 5 | DPP: 8(7)

## Pole and Polar

**Question 118**
Define pole and polar for a 2nd degree curve.

**Model Answer**
Through $P$ draw a variable line meeting the curve at $Q, R$; the locus of the meet of the tangents at $Q, R$ is the polar of $P$, and $P$ is the pole of that line.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 119**
Write the equation of the polar of a given point.

**Model Answer**

$$T = 0$$

Conjugate points: each lies on the polar of the other. Ex: $(2, k)$, $(k, 3)$ conjugate w.r.t. $x^2+y^2=10$: $2k+3k=10$, $k=2$.

**Grade** - Low
Practice map, main concept: none
Also used in: Sheet: C-II 3

**Question 120**
What does T = 0 represent for a point, according to its position relative to the circle?

**Model Answer**
Outside: chord of contact (= polar); on the circle: tangent at the point; inside: polar only (no chord of contact), a line not meeting the circle.

**Grade** - Low
Practice map, main concept: none
Also used in: none

**Question 121**
How to find the coordinates of the pole of a given line?

**Model Answer**
Write $T = 0$ for $(x_1, y_1)$ and compare with the line. For $lx + my = n$ and $x^2 + y^2 = a^2$: pole $\left(\frac{a^2 l}{n}, \frac{a^2 m}{n}\right)$.

**Grade** - Medium
Practice map, main concept: Sheet: J-A 13
Also used in: none

**Question 122**
From any variable point on a fixed line, tangents are drawn to a 2nd degree curve; the chord of contact passes through which point?

**Model Answer**
Through the pole of the fixed line.
Ex: points on $2x + y = 4$, circle $x^2 + y^2 = 1$: chords of contact pass through $(\frac{1}{2}, \frac{1}{4})$.

**Grade** - Low
Practice map, main concept: Sheet: J-II 28
Also used in: none

**Question 123**
What is the locus of the harmonic conjugate of a point P w.r.t. the 2 points in which any variable line through P cuts a 2nd degree curve?

**Model Answer**
The polar of $P$: $T = 0$ (for $x^2 + y^2 = a^2$: $xx_1 + yy_1 = a^2$).

**Grade** - Low
Practice map, main concept: none
Also used in: none